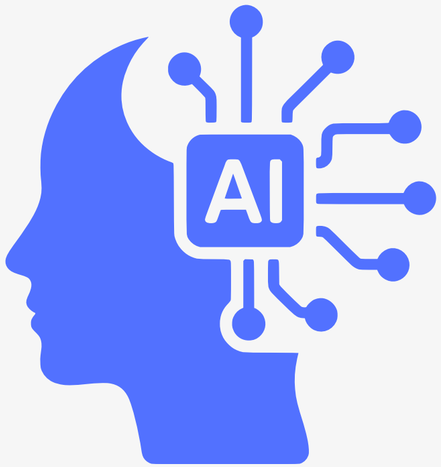
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# PROMPT VERSIONING & AI CI/CD

YOUR AI STARTS GIVING INCORRECT ANSWERS.
HOW DO YOU IDENTIFY THE PROBLEM?

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# WHAT IS PROMPT ENGINEERING?

- Prompt Engineering is the process of designing and improving instructions given to an AI model.
- A good prompt helps an LLM generate more accurate, relevant, and consistent responses.
- Example: Creating a prompt that instructs an AI customer support assistant to answer politely and follow company policies.

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# WHAT IS PROMPT MANAGEMENT?

- Prompt Management means organizing, storing, maintaining, and managing prompts used in AI applications.
- Instead of hardcoding prompts throughout the application, we manage them centrally.
- It makes prompts easier to update, test, reuse, and maintain.

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# WHAT IS PROMPT VERSIONING?

- Prompt Versioning means maintaining different versions of the same prompt.
- Every important prompt change can be tracked as a new version.
- It allows developers to compare versions, identify problems, and roll back to a previous working prompt.

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# WHAT IS CI/CD?

- CI (Continuous Integration): Automatically integrating and checking code changes whenever developers make updates.
- CD (Continuous Delivery/Deployment): Automating the process of preparing and releasing application changes.
- CI/CD helps teams deliver software faster, more consistently, and with fewer manual errors.

# WHAT IS AI CI/CD?

- AI CI/CD extends traditional CI/CD practices to AI-powered applications.
- It automates the validation, testing, and deployment of AI application changes, including prompts and model configurations.
- It helps ensure that updates do not break existing AI functionality.

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# WHAT ARE AI DEPLOYMENT WORKFLOWS?

- Deployment workflows define how AI application changes move from development to production.
- They include prompt updates, validation, testing, approval, deployment, and monitoring.
- Example workflow: Developer updates a prompt → tests run → changes get approved → updated prompt is deployed.

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# WHAT ARE TESTING PIPELINES?

- A testing pipeline is an automated sequence of tests executed before changes are released.
- It checks whether the updated prompt or AI application behaves as expected.
- Testing pipelines help detect problems before users experience them.

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# IMPORTANT AI TESTING TERMINOLOGIES

- Prompt Testing: Checking whether a prompt produces the expected responses.
- Regression Testing: Ensuring that a new prompt version does not break previously working functionality.
- Evaluation (Evals): Measuring AI output quality against defined criteria.
- A/B Testing: Comparing two prompt versions to identify which performs better.
- Validation: Checking whether changes meet predefined requirements before deployment.
- Rollback: Reverting to a previously working prompt or application version when an update causes problems.

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# WHAT IS A PROMPT DEPLOYMENT WORKFLOW?

- A typical workflow looks like this:
- Create: Write or update a prompt.
- Version: Save the prompt as a new version.
- Test: Run automated prompt tests.
- Evaluate: Check accuracy, relevance, and response quality.
- Approve: Review and approve the changes.
- Deploy: Release the approved prompt to production.
- Monitor: Observe its performance and identify issues.
- Rollback: Restore a previous version if something goes wrong.

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# WHY DO WE NEED PROMPT VERSIONING & AI CI/CD?

- To prevent unexpected prompt changes from breaking AI applications.
- To maintain consistent AI response quality.
- To automate testing and deployment.
- To track changes and collaborate with other developers.
- To safely release improvements and quickly recover from failures.